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DALI-SLAM: Degeneracy-aware LiDAR-inertial SLAM with novel distortion correction and accurate multi-constraint pose graph optimization

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ABSTRACT

LiDAR-Inertial simultaneous localization and mapping (LI-SLAM) plays a crucial role in various applications such as robot localization and low-cost 3D mapping. However, factors including inaccurate motion distortion estimation and pose graph constraints, and frequent LiDAR feature degeneracy present significant challenges for existing LI-SLAM methods. To address these issues, we propose DALI-SLAM, an accurate and robust LI-SLAM that consists of degeneracy-aware LiDAR-inertial odometry (DA-LIO) with a dual spline-based motion distortion correction (DS-MDC) module, and multi-constraint pose graph optimization (MC-PGO). Considering the cumulative errors of micro-electromechanical systems (MEMS) inertial measurement unit (IMU) integration, two continuous-time trajectories in the sliding window are fitted to update the discrete IMU poses for accurate motion distortion correction. In the LiDAR-inertial fusion stage, LiDAR feature degeneracy is detected by analyzing the Jacobian matrix and a remapping strategy is introduced into the updating of error state Kalman Filter (ESKF) to mitigate the influence of degeneracy. Furthermore, in the back-end optimization stage, three types of submap constraints are accurately built with dedicated strategy through a robust variant of the iterative closest point (ICP) method. The proposed method is comprehensively validated using data collected from a helmet-based laser scanning system (HLS) in representative indoor and outdoor environments. Experiment results demonstrate that the proposed method outperforms the SOTA methods on the test data. Specifically, the proposed DS-MDC module reduces trajectory root mean square errors (RMSEs) by 7.9 %, 5.8 %, and 3.1 %, while the degeneracy-aware update strategy achieves additional reductions of 43.3 %, 17.7%, and 4.9 %, respectively, across three typical sequences compared to existing methods, thereby effectively improving trajectory accuracy. Furthermore, the results of DA-LIO demonstrate a maximum RMSE within 1 kilometer of approximately 1 meter in outdoor environments, achieving superior performance compared to the SOTA method FAST-LIO2. After performing MC-PGO, the RMSEs of the trajectories are reduced by 25.2 %, 9.2 %, and 52.4 %, respectively, across three typical sequences, demonstrating better performance compared to the SOTA method HBA. Code will be available at https://github.com/DCSI2022/DALI_SLAM.

1. Introduction

Simultaneous localization and mapping (SLAM), since introduced by Smith and Cheeseman (1986), has made significant progress after decades of development (Cadena et al., 2016), and is applied in various fields, including robot localization and navigation (Tranzatto et al., 2022), augmented reality/virtual reality (AR/VR) (Davison et al., 2007; Li et al., 2024a) and mobile mapping (Wu et al., 2019; Shao et al., 2020; Karam et al., 2021). Based on the sensor type, the SLAM methods can be categorized into visual SLAM (Mur-Artal et al., 2015), LiDAR SLAM (Pan et al., 2024), and multi-sensor fusion-based SLAM (Lin and Zhang, 2022). LiDAR sensor can directly capture a high-precision point cloud of the environment and is less affected by illumination. With the development of solid-state LiDAR technology (Liu et al., 2021; Zhang et al.,

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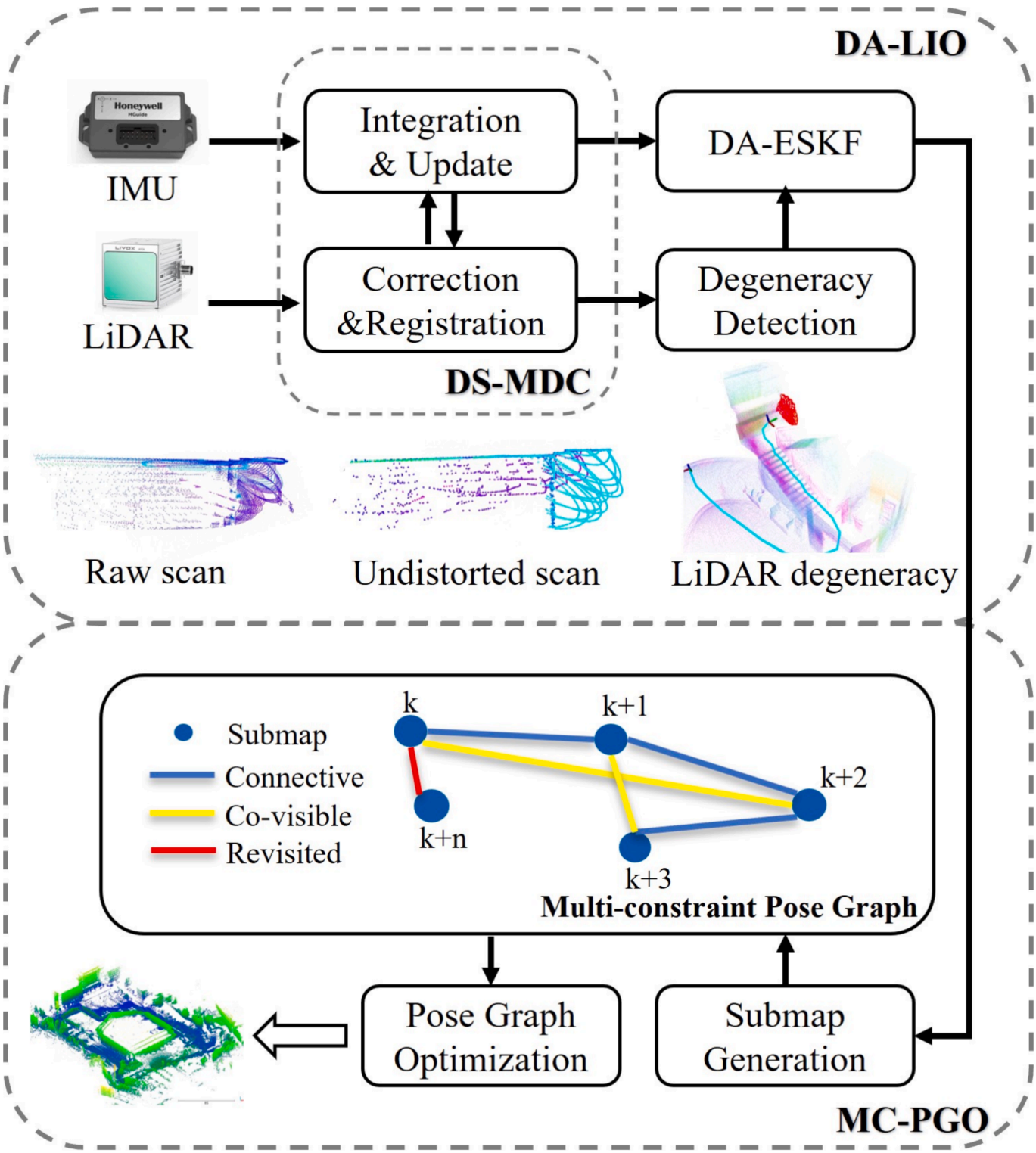


Fig. 1. The workflow of DALI-SLAM. The degeneracy-aware LiDAR-inertial odometry (DA-LIO) which includes the dual spline-based point cloud motion distortion correction (DS-MDC) module and a degeneracy-aware ESFK module, first undistorts the point cloud of each frame and then estimates the system's pose by utilizing consecutive LiDAR and IMU measurements. The multi-constraint pose graph optimization (MC-PGO) first generates point cloud submaps and then constructs a pose graph with three types of constraints to optimize submap poses, ensuring globally consistent mapping.

2022), LiDAR sensors have shown promising application prospects due to characteristics such as small size, high reliability, and low cost.

A LiDAR SLAM system typically comprises two main parts: a front-end that performs LiDAR odometry (LO) and a back-end optimization module. Scan matching, the process of finding the optimal transformation to align two point clouds, is the core of LO (Pomerleau et al., 2015). Significant efforts have been made by researchers to enhance the accuracy of LO through various aspects with respect to scan matching, including trajectory modeling (Wu et al., 2023), feature selection (Jiao et al., 2021), and outlier rejection (Chebrolu et al., 2021), among others. However, LO performs poorly in situations with rapid motion and when LiDAR feature degeneracy occurs. The fusion of data from the LiDAR and

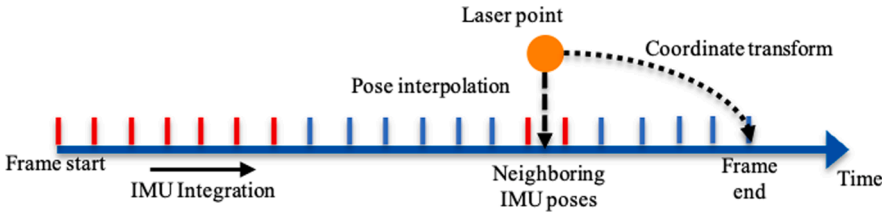


Fig. 2. Motion distortion correction using IMU integration. The relative pose of each laser point with respect to the coordinate at the start of the frame is calculated by performing linear interpolation between neighboring IMU poses, after which the laser point is transformed to the coordinate at the end of the frame.

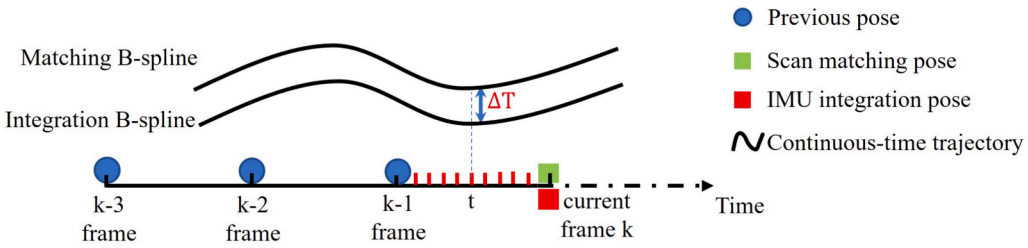
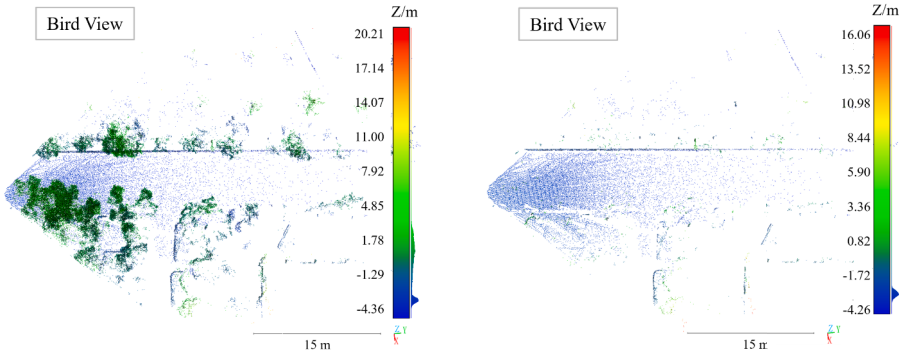


Fig. 3. IMU pose update through fitting continuous-time trajectory. A matching B-spline is fitted utilizing previous poses along with the current frame pose estimated by scan matching; while an integration B-spline is fitted utilizing previous poses along with the current frame pose derived from IMU integration. Each IMU pose is then updated by the difference between the two trajectories.



(a) raw submap (b) submap with non-planar points filtered

Fig. 4. Preprocessing of non-planar point filtering prior to submap registration. (a) shows the raw point clouds of the submap in the test data. (b) displays the submap after filtering, retaining planar points on surfaces like the ground and tree trunks.

the inertial measurement unit (IMU) is complementary, as the IMU sensor provides high frequency angular velocity and acceleration measurements of the platform and renders gravity observable.

Although LiDAR-inertial odometry (LIO) has attracted wide attention in recent years (Xu et al., 2022; Li et al., 2024b), there are still some limitations. Firstly, motion distortion in the point cloud within a frame significantly decreases the scan matching performance as it breaks the rigid transformation assumption. Discrete-time trajectory methods (Xu and Zhang, 2021) typically rely on integrated IMU poses to correct motion distortion. However, the inherent cumulative errors of micro-electromechanical systems (MEMS) IMU integration are neglected, leading to larger errors in points towards the end of each frame. Secondly, LiDAR feature degeneracy refers to the phenomenon of 'sliding' or complete pose unobservability in scan matching caused by the lack of geometric features and uneven distribution of features (Gelfand et al., 2003). Solid-state LiDAR sensors, such as the Livox Avia, are more susceptible to degeneracy than mechanical LiDARs due to their lower measurement rate and limited horizontal field of view (FOV), which result in fewer feature points per scan. However, the influence of degenerate LiDAR matching information is not considered in existing tightly-coupled LIO systems. Lastly, compared to LIO, the back-end optimization is equally important for reducing front-end drift and improving geometric consistency of the map. However, there is limited research on practical strategies for constructing accurate pose graph constraints, especially for LiDAR with limited FOV.

To address the aforementioned issues, the objective of this paper is to improve the accuracy and robustness of LI-SLAM methods by integrating several novel modules into the front-end and back-end. We propose a degeneracy-aware tightly-coupled LiDAR-inertial SLAM method, which achieves efficient and accurate localization and mapping. The contributions of this paper are as follows:

- A novel dual spline-based point cloud motion distortion correction (DS-MDC) module that considers IMU integration errors is proposed and integrated into a filter-based LIO system. It utilizes the registration pose of the current frame to the map and the integration pose of IMU to separately fit continuous-time trajectories for updating the discrete IMU poses. Compared to the existing methods, the RMSEs of the trajectories are reduced by 7.9 %, 5.8 % and 3.1 % respectively.
- To mitigate the impact of LiDAR degeneracy on sensor fusion, a degeneracy-aware Kalman filter update strategy is proposed by first introducing a Jacobian remapping algorithm into the tightly-coupled

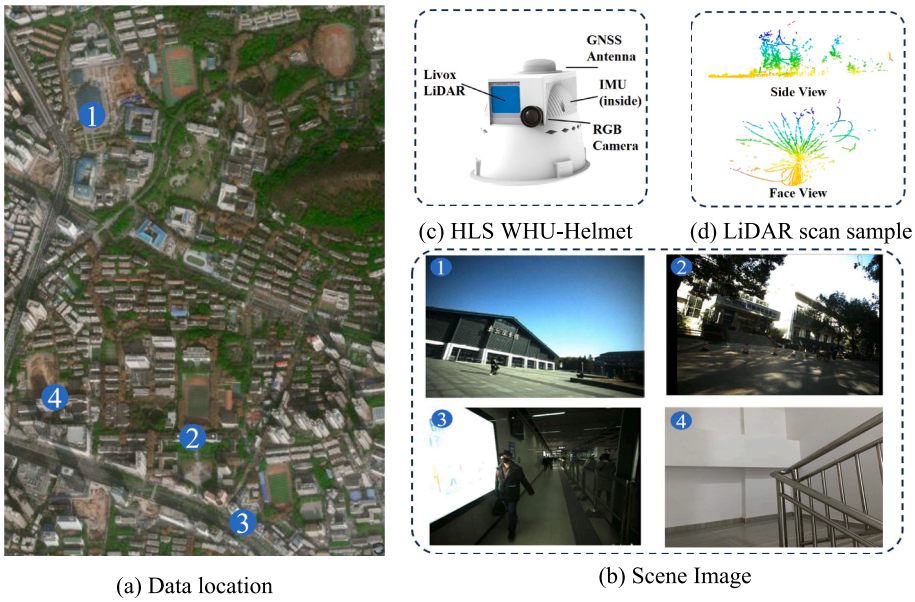


Fig. 5. Overview of experiment data.

Table 1
The detailed description of experiment data.

Dataset	Description				
	Scenario	Trajectory length/m	Duration/ seconds	LiDARframe num	LiDAR type
Seq1	Campus Road	896	661	6610	Mid-70
Seq2	Campus Road	1201	841	8410	Mid-70
Seq3	Subway Station	282	180	1800	Avia
Seq4	Staircase	25	50	500	Avia

Table 2
Front-end results of different methods on experiment data.

Dataset	Method		
	CLINS ATE RMSE/Max error (m)	FAST-LIO2	DA-LIO (Ours)
Seq1	0.897/2.082	1.051/1.840	**0.484/1.017**
Seq2	1.771/3.605	1.068/2.182	**0.693/1.434**
Seq3	Failed	2.579/5.298	**2.376/4.850**

filter-based LIO system. Compared to the existing strategy, the RMSE of the trajectories are reduced by 43.3 %, 17.7 % and 4.9 % respectively.

- We propose an accurate and robust LI-SLAM method that consists of two parts: degeneracy-aware LiDAR-inertial odometry (DA-LIO) and multi-constraint pose graph optimization (MC-PGO). Different types of relative pose constraints are accurately built through a robust variant of iterative closest point (ICP) (Pomerleau et al., 2015) method with dedicated strategy. The method undergoes comprehensive validation using data acquired by the WHU-Helmet system in various indoor and outdoor scenarios. The normalized trajectory RMSEs of our method are 0.4 ‰, 0.52 ‰, and 4.01 ‰ respectively on the test data.

The rest of the paper is organized as follows. Section 2 reviews the related work. Section 3 introduces the proposed DALI-SLAM method. Section 4 presents the experiment data and results. Section 5 presents the comparative study and discusses the deficiency of the method. Section 6 concludes the paper.

2. Related work

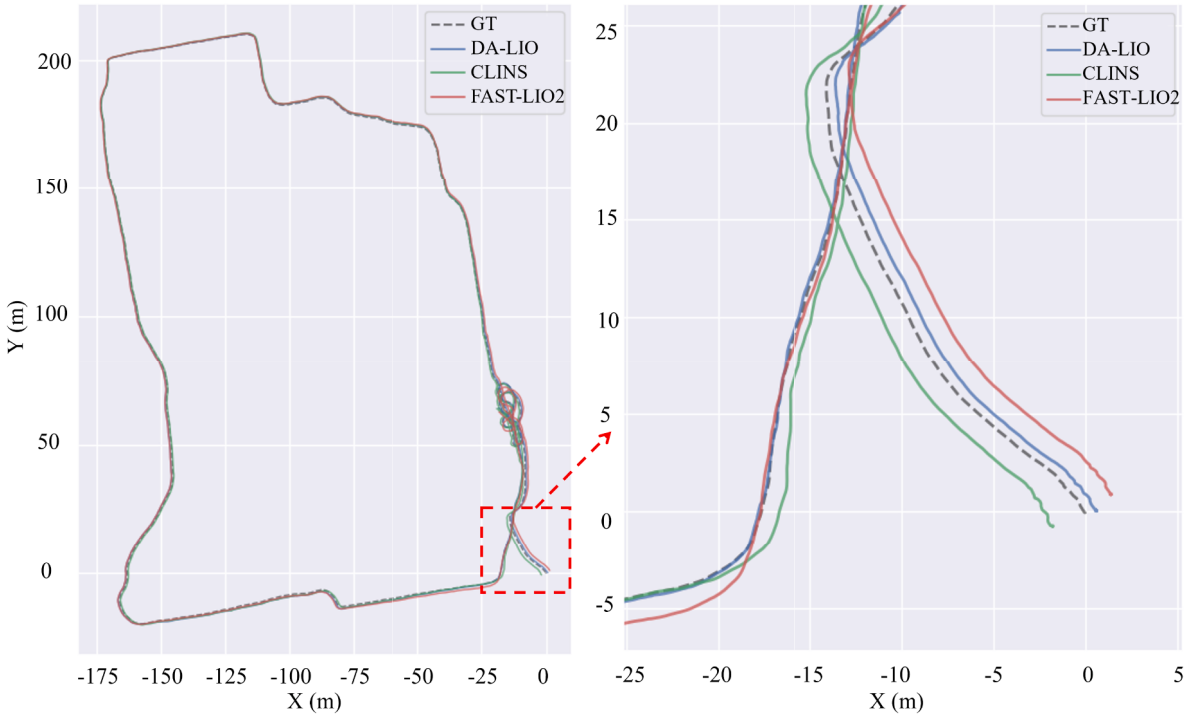
In this section, we review the related work in terms of LiDAR degeneracy detection and handling, motion distortion correction and point cloud map optimization.

2.1. LiDAR degeneracy detection and handling

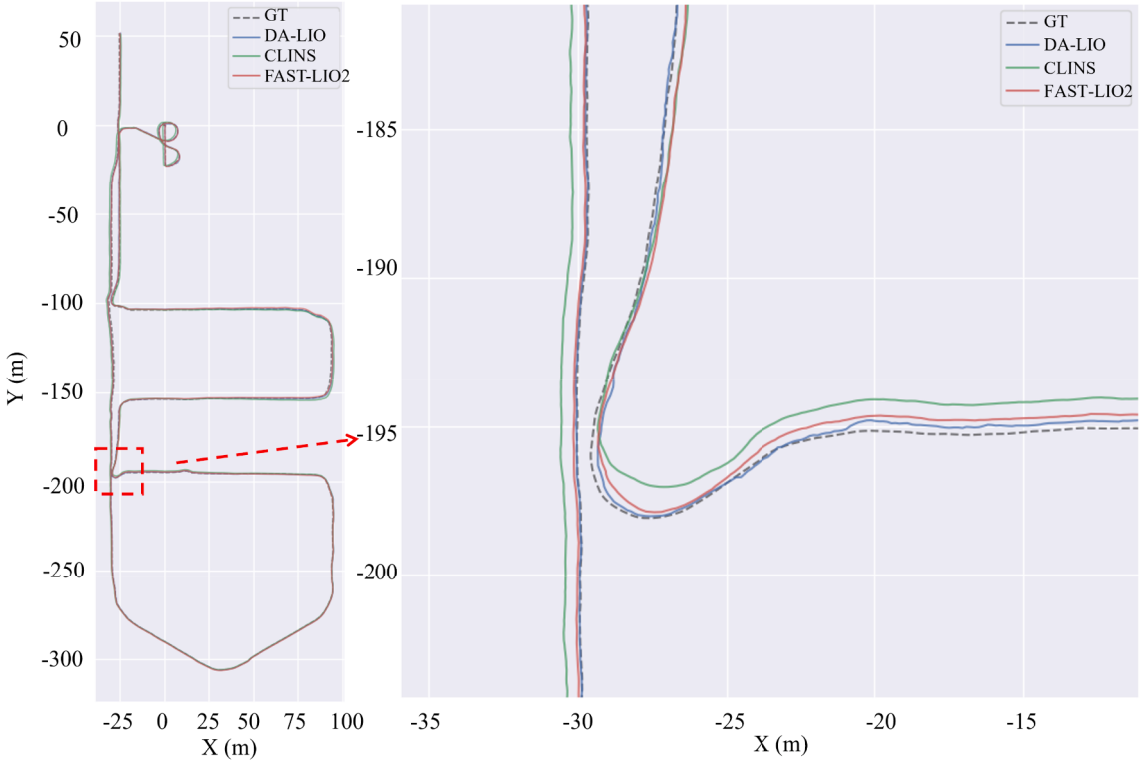
The methods for the LiDAR degeneracy detection can be classified into four categories: sampling-based, optimization-based, geometry-based, and learning-based. For the sampling-based approach, the unscented transform with a limited number of sigma points were introduced to accurately sample the 3D ICP covariance (Brossard et al., 2020), which, to some extent, reflects the degree of LiDAR degeneracy. For the optimization-based approach, Zhang et al. (2016) proposed an efficient and simple degeneracy detection method, where the degeneracy factor is derived from the minimum eigenvalue of the optimization system. Through analyzing the contribution of geometric correspondences to the principle directions of the ICP optimization, Tuna et al.(2023) proposed a robust fine-grained localizability detection module. Nubert et al.(2022) pioneered the use of 3D convolutional neural networks to infer the degeneracy of single-frame point cloud in six degree of freedom (DOF) directions. However, this method is limited to scan-to-scan point cloud registration.

Multi-sensor fusion strategy is important for robust and accurate pose estimation when LiDAR degeneracy occurs (Liao et al., 2024). Zhang and Singh (2018) utilized the solution remapping algorithm (Zhang et al., 2016) to update the state when the degeneracy is detected in the sequential processing pipeline for LiDAR-visual-inertial odometry. In the loosely-coupled system, Wu et al., 2022a employed an information management module to select non-degenerate subsystem (e.g., visual-inertial system) to update the system state. To eliminate the effect of LiDAR degeneracy in the tightly-coupled optimization, Wen et al. (2024) associated the LiDAR measurement residuals only with the non-degenerate DOF of the states, which were detected with the assistance of visual-inertial odometry.

In summary, for LiDAR degeneracy detection, the sampling-based methods tend to have relatively low computational efficiency. The

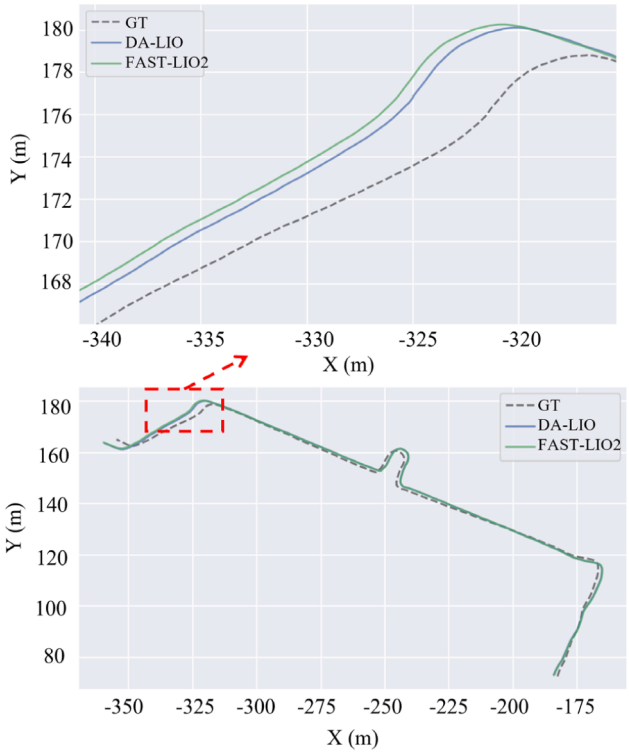


(a) Seq1



(b) Seq2

Fig. 6. Trajectory of different method in x-y projection on experiment data.



(c) Seq3

Fig. 6. (*continued*).

optimization-based methods are relatively simple and efficient. The geometric-based methods require careful analysis of the geometric information to correctly identify the degeneracy directions. As for the learning-based methods, they still need to demonstrate their generalization capabilities across a broader range of scenarios and different LiDAR sensor types. To better utilize raw LiDAR and IMU data and mitigate the impact of LiDAR degeneracy on sensor fusion, we firstly leverage an optimization-based LiDAR degeneracy detection method and secondly introduce a Jacobian remapping strategy in the tightly-coupled filter-based LIO system.

2.2. Motion distortion correction

The existing methods can be classified into two categories based on how they integrate MDC with pose estimation.

One category of research leverages continuous-time (CT) models to simultaneously estimate the trajectory and correct motion distortion in LiDAR point cloud. Zheng and Zhu (2024) proposed Traj-LO, which achieves performance comparable to LIO methods even in scenarios involving rapid or extreme motion by utilizing a linear CT trajectory and incorporating smoothness constraints to improve trajectory estimation accuracy and robustness. However, a fixed frame length cannot guarantee the linearity of motion within the frame. To address this limitation, Wu et al.(2023) proposed the adaptive frame length LO to reduce the impact of non-linear motion on scan matching based on linear CT trajectory model. These methods utilize a linear CT model, which allows for motion distortion correction without requiring IMU data. Lv et al. (2021) introduced a B-spline model to rigorously fuse asynchronous LiDAR and IMU measurements. However, the optimal selection of spline knot spacing remains an unsolved problem (Talbot et al., 2024). Alternatively, Burnett et al. (2024) employed a Gaussian process motion prior to estimate the CT trajectory for LiDAR-inertial systems.

The other category of work utilizes prior motion model (e.g., constant velocity) or other sources of poses to correct distortion before pose estimation. Zhu et al. (2022) adopted a constant velocity model to compensate the distortion in a LiDAR scan without the aid of other sensors. Additionally, IMU poses derived by integration is commonly used for correction. To get the relative pose of the points within a frame, linear interpolation is often performed between neighboring IMU poses. However, when facing intense motion, the linear interpolation errors cannot be neglected. Le Gentil et al.(2018) employed upsampling of IMU measurements to tackle the issue. In addition, Chen et al.(2023) proposed using the constant acceleration assumption to model the motion between adjacent IMU poses at a higher order, enabling the capture of higher-frequency movements. Considering the trajectory uncertainty of IMU, Deschênes et al.(2021) proposed different point weighting strategies in scan matching under extreme motions.

In summary, the cumulative errors of IMU integration are neglected in existing distortion correction methods. We utilize the CT model to update high-frequency IMU poses within the duration of one LiDAR frame, instead of directly optimizing the CT model for pose estimation, thereby preventing over-fitting or under-fitting of the trajectory. Specifically, we derive the correction of IMU poses through comparing two B-spline trajectories for more accurate MDC.

2.3. Point cloud map optimization

Due to the registration error between point cloud frames and map, odometry drift occurs and further leads to the geometric inconsistency of point cloud map. Existing methods for point cloud map optimization can be mainly categorized into PGO and LiDAR bundle adjustment (BA).

The PGO methods indirectly optimize the geometric structure of map by minimizing the relative pose errors between graph nodes efficiently (Nüchter et al., 2007). Mendes et al.(2016) proposed a pose graph SLAM method for the mechanical LiDAR. This method detects potential loops between key frames, and performs ICP registration between the key

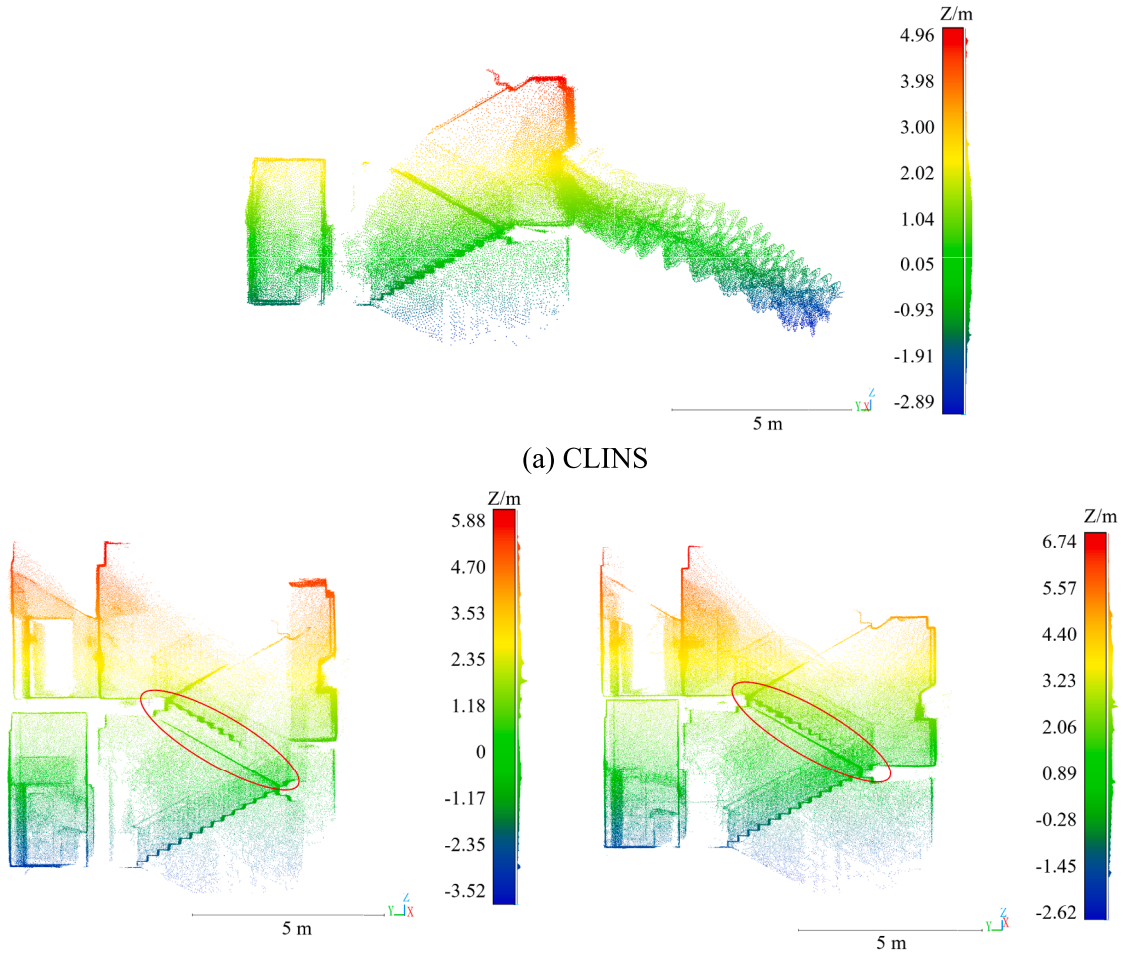


Fig. 7. point clouds of different LIO methods on Seq4.

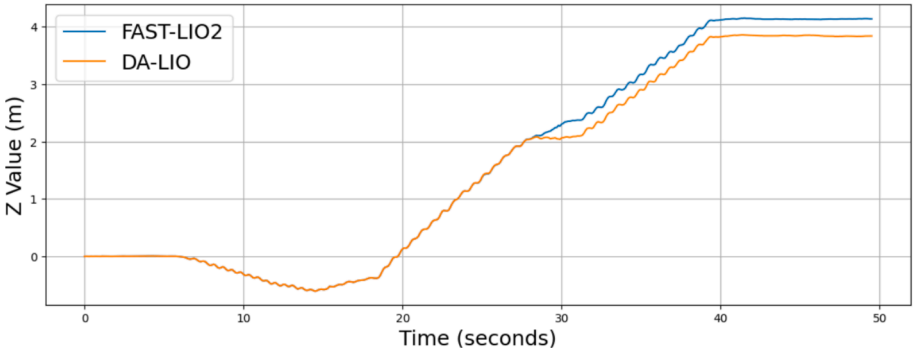


Fig. 8. Z-Axis trajectory of different LIO methods on staircase dataset.

Table 3
Evaluation of the efficiency of different LIO methods.

Metrics	CLINS	FAST-LIO2	DA-LIO (Ours)
Average time (ms)	93.3	**16.9**	42.8
STD of time (ms)	35.8	**7.5**	14.1

frame and the submap to construct relative pose constraints. To establish the correct correspondences for revisited areas in the mobile laser scanning point clouds, Li et al., 2023b introduced multi-scale constrains into the pose graph for correcting position inconsistency. Recently, Zou et al.(2024) proposed an long-term association mapping framework that dynamically reloads map points after pose graph optimization. However, to accurately obtain the relative pose constraints between nodes, these methods impose high adaptability requirements on point cloud registration.

In recent years, there has been a gradual increase in research on LiDAR BA which directly optimize the map structure. Liu and Zhang (2021) proposed a local BA method for LiDAR mapping where plane and edge features parameters can be analytically solved, reducing the

Table 4
Back-end results of different methods on experiment data.

Dataset	Method				
	DA-LIO	LeGO-LOAM-B	HBA	MuCoGraph	MC-PGO (Ours)
	ATE RMSE/ Max error (m)				
Seq1	0.484/1.017	1.111/ 2.194	0.365/ 0.928	0.419/ 0.969	**0.362/ 0.762**
Seq2	0.693/1.434	0.693/ 1.434	0.641/ 1.405	0.697/ 1.461	**0.629/ 1.386**
Seq3	2.376/4.850	2.376/ 4.850	2.350/ 4.870	2.376/ 4.850	**1.130/ 2.622**

Table 5
The number of different submap constraints on experiment data.

	Submap	Connective	Revisited	Co-visible
Seq1	132	83	9	26
Seq2	120	88	4	14
Seq3	90	89	0	273

dimension of the optimization. Subsequently, Liu et al. (2023b) proposed encoding all points associated with the same feature into point cluster coordinates, which eliminates the need to enumerate each point for derivative evaluation and thus improves efficiency. High-level geometric primitives (e.g., plane) need to be accurately extracted for data association. Zhou et al. (2021) addressed double-side issue that may cause wrong data association in the plane adjustment. Liu et al. (2023a) proposed a consistent mapping framework that includes bottom-up hierarchical LiDAR BA and top-down PGO, improving the optimization efficiency of such methods in large-scale scenes.

Compared to the mechanical LiDAR, solid-state LiDAR generally has a narrower horizontal FOV. This characteristic leads to a reduced number of effective feature points captured in a single scan, rendering SLAM algorithms more vulnerable to environmental influences, especially for the point cloud map optimization. Therefore, point cloud accumulated by a period of time is chosen as the basic unit for PGO. In addition, a robust variant ICP with dedicated strategy is proposed to build accurate relative pose constraints between graph nodes.

3. Methodology

3.1. Overview

Our method (DALI-SLAM) follows the consistent design of SLAM approaches in terms of architecture, as it consists of two modules: the front-end (DA-LIO) and the back-end (MC-PGO), shown as Fig. 1. The system receives synchronized LiDAR scans and IMU data, and the pose is generated at the same frequency as the IMU data, while simultaneously outputting a global consistent map.

In the front-end, the raw scan data is firstly corrected through a DS-MDC module. We propose a degeneracy-aware ESKF framework to fuse the LiDAR and IMU measurements. Specifically, when degeneracy is detected, the Jacobian matrix of the LiDAR matching is remapped to update the ESKF. In the back-end, submaps that accumulate point cloud over a short period (e.g., 5 s) are firstly generated using the output from the DA-LIO. Then, a multi-constraint pose graph is built and optimized to further decrease the front-end errors.

3.2. DA-LIO

To efficiently fuse LiDAR scan measurements with high-frequency IMU measurements, the IMU measurements between adjacent scans are integrated, and the system state $\boldsymbol{x}$ is updated at the frequency of the LiDAR scan through iterated error state Kalman filter (EKSF) (Xu et al., 2022). The state vector in DA-LIO is defined as:

$$\boldsymbol{x} = \left[\boldsymbol{p}_I^W, \boldsymbol{R}_I^W, \boldsymbol{v}^W, \boldsymbol{b}_a, \boldsymbol{b}_g, \boldsymbol{g}^W\right]^{\mathrm{T}}. \tag{1}$$

$\boldsymbol{x}$ exists on a high-dimensional manifold, where $\boldsymbol{p}, \boldsymbol{v} \in \mathrm{R}^3$ represents the position and velocity, $\boldsymbol{R} \in \mathrm{SO}(3)$ denotes the rotation, $\boldsymbol{b}_g, \boldsymbol{b}_a$ represent the IMU biases, and $\boldsymbol{g}^w$ represents gravity vector. The position and rotation are defined as the transformation vector or matrix from the body coordinate system I to the world coordinate system W. Bold variables are used to represent vectors or matrices

However, the errors accumulated during IMU integration can propagate to the motion distortion correction of the LiDAR scan, thereby affecting the accuracy of LIO. In addition, LiDAR feature degeneracy can lead to erroneous state estimation. Next, we will focus on enhancing the performance of LIO based on the ESKF framework in two key aspects, which will be detailed in Sections 3.2.1 and 3.2.2.

3.2.1. DS-MDC

Errors arising from the process of correcting motion distortion (as shown in Fig. 2) in a single-frame point cloud through IMU integration (Forster et al., 2015), pose interpolation, and coordinate transformation, are generally considered negligible. In fact, both the accuracy of interpolation and the integration affect the quality of the correction. In this subsection, we aim to enhance the accuracy of MDC by focusing on updating IMU poses.

The core idea is to leverage the low-frequency scan matching poses to correct the high-frequency IMU poses, with a continuous-time trajectory employed to bridge the frequency gap. As shown in Fig. 3, we utilize the updated poses of the previous frames from the ESKF and different current frame poses to fit the integration B-spline and the matching B-spline respectively. The difference ΔT between the two splines is then used to update the IMU poses $\boldsymbol{T}_t^I$:

$$\overline{\boldsymbol{T}_t^I} = \mathrm{Spline}_{\mathrm{match}}(t) \bullet \mathrm{Spline}_{\mathrm{integration}}(t)^{\mathrm{T}} \bullet \boldsymbol{T}_t^I, \tag{2}$$

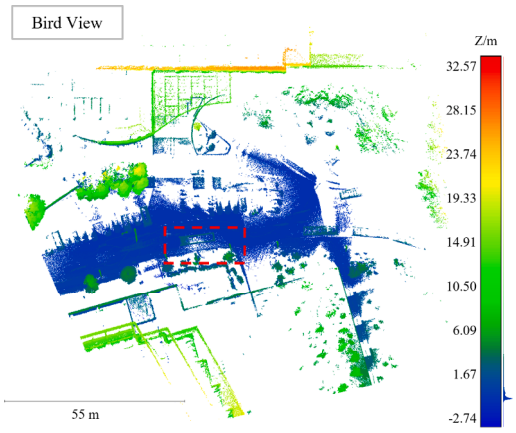
where $\mathrm{Spline}(t)$ represents the 6 DOF pose of the spline at the time t, $\boldsymbol{T}_t^I$ represents the pose derived by IMU integration at the time t. We employ a split B-spline representation that separates position and orientation components. More details of fitting B-splines can be found in Sommer et al.(2020). Algorithm 1 outlines the proposed DS-MDC algorithm, incorporating the IMU pose update. In addition, the scan matching pose is obtained by aligning the current frame, which has been corrected by initial IMU poses, with the map. Specific scan matching method will be further described in section 3.3.

Algorithm 1 The Dual Spline-Based Motion Distortion Correction

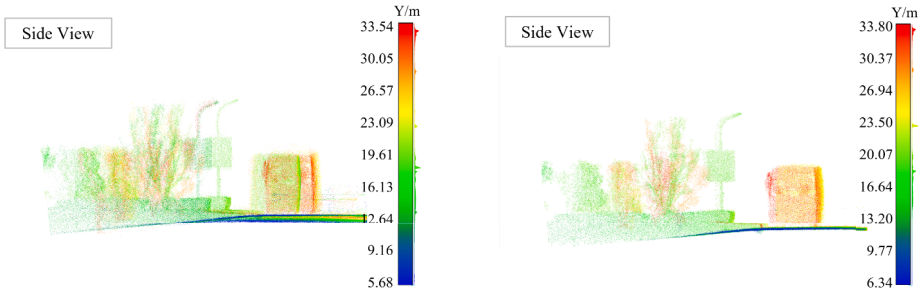
Input: raw points of one LiDAR frame $\{\boldsymbol{p}\}$, Map points $\mathcal{M}$, IMU poses within current frame $\{\boldsymbol{T}_t^I\}$, previous frame poses $\{\boldsymbol{T}_k^I\}$;
Output: corrected points of one LiDAR frame $\{\boldsymbol{p}'\}$;
1. **MDC** $(\{\boldsymbol{p}\}, \{\boldsymbol{T}_t^I\}) \rightarrow$ Obtain $\{\boldsymbol{p}''\}$ // Motion distortion correction using IMU poses as shown in Fig. 2;
2. **Scan Matching** $(\{\boldsymbol{p}''\}, \mathcal{M}, \boldsymbol{T}_{end}^I) \rightarrow$ obtain end-time pose of current frame $\boldsymbol{T}_{end}^p$// Using $\boldsymbol{T}_{end}^I$ as the initial value for scan matching; refer to Section 3.3, Equation 5;
3. Fit **integration B-spline** using $\{\boldsymbol{T}_k^I\}$ and the end-time pose of current frame $\boldsymbol{T}_{end}^I$ derived from IMU integration;
4. Fit **matching B-spline** using $\{\boldsymbol{T}_k^I\}$ and the end-time pose of current frame $\boldsymbol{T}_{end}^p$ derived from scan matching;
5. Update IMU poses using Equation 2 to obtain $\{\overline{\boldsymbol{T}_t^I}\}$;
6. **MDC** $(\{\boldsymbol{p}''\}, \{\overline{\boldsymbol{T}_t^I}\}) \rightarrow$ Obtain $\{\boldsymbol{p}'\}$; // Motion distortion correction using updated IMU poses.

3.2.2. DA-ESKF

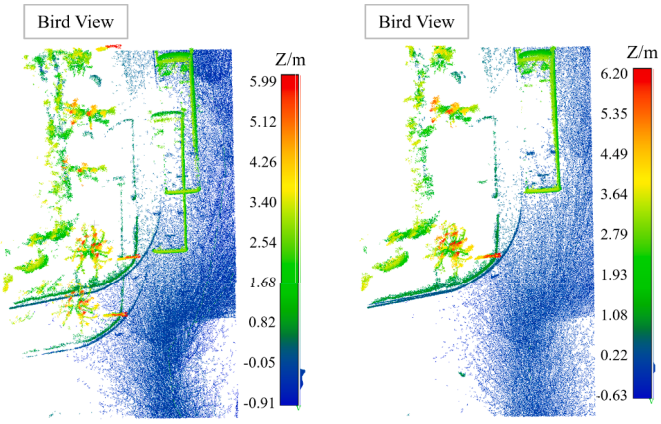
The LIO method based on the ESKF framework has demonstrated superior performance in feature-rich environments. However, existing update strategies of Kalman filter can lead to inaccurate state estimation in scenarios with degenerate scene features. Building upon the research



(a) Looped region on Seq1



(b) Side view comparison of point cloud before (left) and after (right) optimization, focusing on the red dashed region marked in (a)



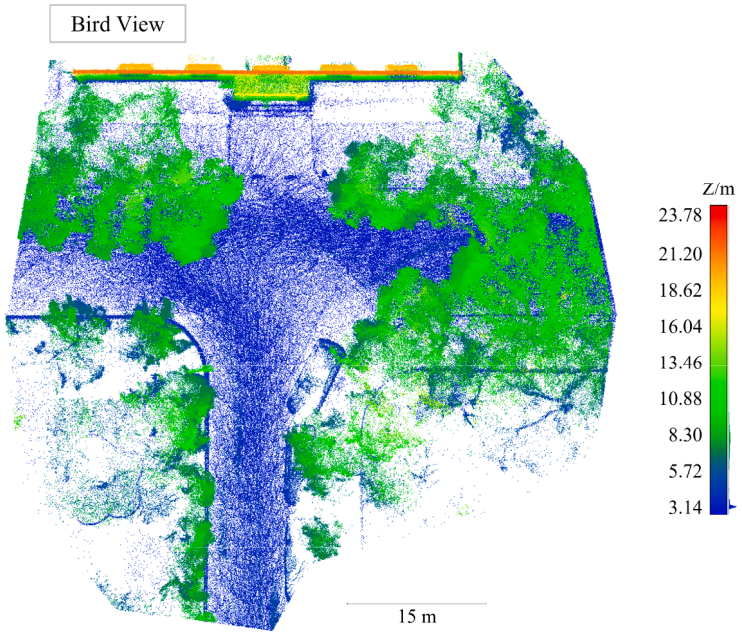
(c) Bird's-eye view comparison of point cloud before (left) and after (right) optimization, focusing on the red dashed region marked in (a)

Fig. 9. Point clouds before (left) and after (right) performing MC-PGO of the looped region on Seq1.

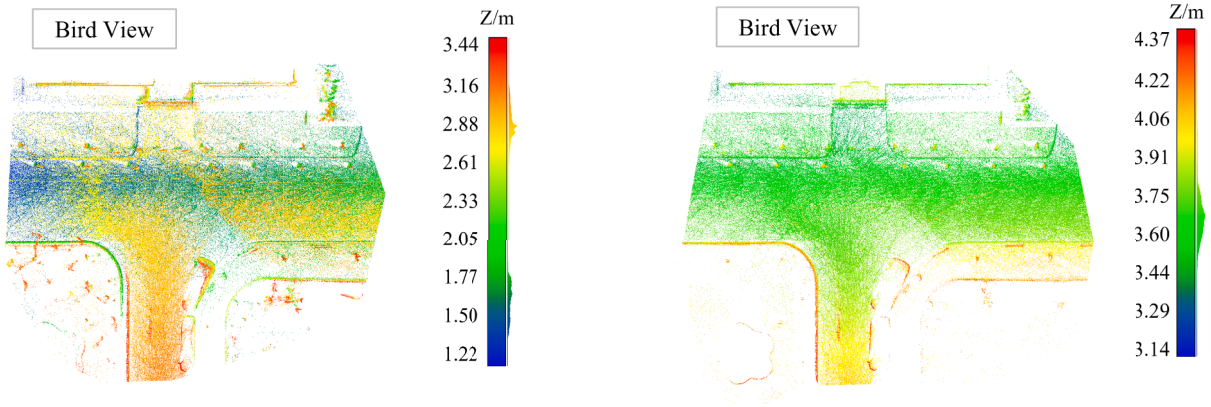
of Zhang et al. (2016) on detecting LiDAR feature degeneracy, we attempt to remap the Jacobian matrix $\boldsymbol{H}$ of LiDAR observations during the filter update stage based on the extent of scene degeneracy. Algorithm 2 describes the Jacobian remapping algorithm. To detect degeneracy, we firstly calculate the eigenvalues of $\boldsymbol{H}^{\mathrm{T}}\boldsymbol{H}$ instead of directly performing singular value decomposition (SVD) on $\boldsymbol{H}$, which can significantly improve the computational efficiency. Degenerate feature directions are detected by comparing eigenvalues with λ_D. If the smallest eigenvalue is less than λ_D, degeneration is detected, and the LiDAR Jacobian matrix undergoes SVD decomposition. If any singular value is smaller than $\sqrt{\lambda_D}$, it is set to a very small value (to ensure numerical stability in matrix computations), and the LiDAR Jacobian matrix is then recalculated.

Algorithm 2 LiDAR Jacobian Matrix Remapping

Input: LiDAR Jacobian matrix $\boldsymbol{H}$;
Output: Jacobian remapping matrix $\boldsymbol{H}'$;
$\boldsymbol{C} = \boldsymbol{H}^{\mathrm{T}}\boldsymbol{H}$;
Eigen Decomposition ($\boldsymbol{C}$), get minimum eigen value λ_1;
If $\lambda_1 < \lambda_D$
SVD($\boldsymbol{H}$), get $\boldsymbol{H} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^{\mathrm{T}}$;
for σ_i in Diag ($\boldsymbol{S}$)
if $\sigma_i < \sqrt{\lambda_D}$
$\sigma_i = 0.0001$; // ensure numerical stability
end for
$\boldsymbol{H}' = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^{\mathrm{T}}$;
else
$\boldsymbol{H}' = \boldsymbol{H}$;



(a) Looped region on Seq2



(b) Bird's-eye comparison of point cloud before (left) and after (right) optimization, focusing on the ground region in (a). Before optimization, the ground exhibits two main elevation distributions, indicating a layered structure. After optimization, geometric consistency was significantly improved.

Fig. 10. Point clouds before and after performing MC-PGO of the looped region on Seq2.

Through the remapping of $\boldsymbol{H}$, the calculation of the Kalman gain $\boldsymbol{K}$ (Xu and Zhang, 2021) is influenced, which restricts the state updates to only in the well-constrained directions derived by LiDAR matchings.

$$K = (\boldsymbol{H}^{\mathrm{T}}\boldsymbol{V}^{-1}\boldsymbol{H}' + \boldsymbol{P}_{pred}^{-1})^{-1}\boldsymbol{H}^{\mathrm{T}}\boldsymbol{V}^{-1}, \tag{1}$$

where $\boldsymbol{P}_{pred}$ represents the predicted state covariance, and $\boldsymbol{V}$ represents the covariance of the LiDAR observation. We adopt the point-to-plane constraint from LOAM (Zhang and Singh, 2014). Consequently, the raw LiDAR Jacobian matrix can be expressed as follows:

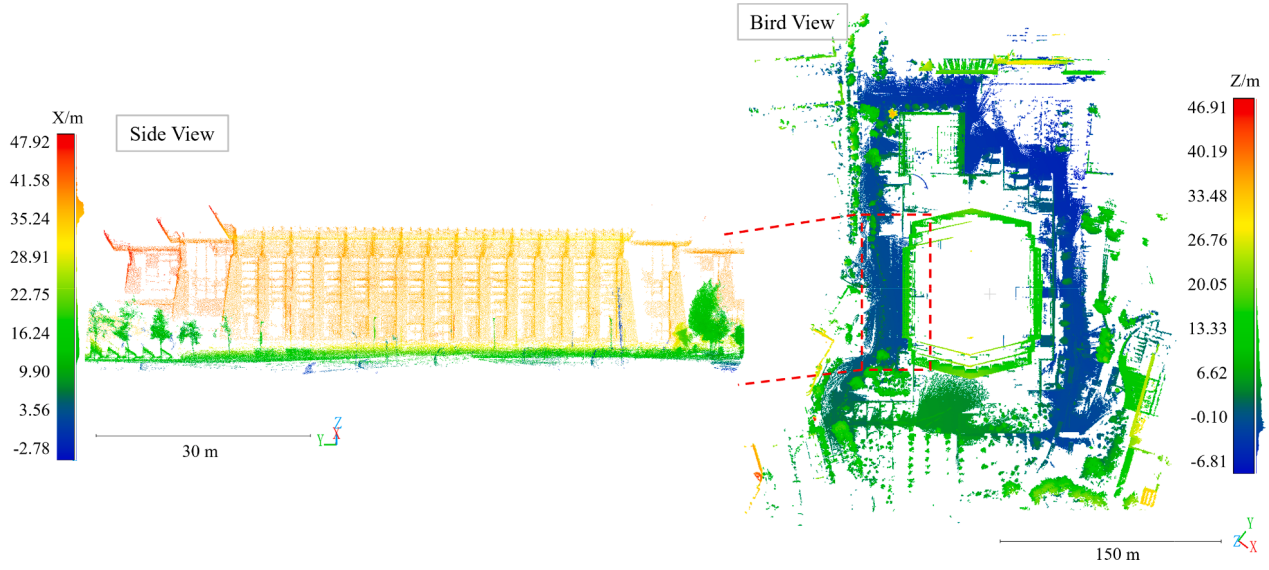
$$H = \underbrace{\begin{bmatrix} \boldsymbol{n}_1 & \boldsymbol{p}_1^W \bullet \boldsymbol{R}^{\mathrm{T}} \bullet \boldsymbol{n}_1 \\ \cdots & \cdots \\ \boldsymbol{n}_m & \boldsymbol{p}_m^W \bullet \boldsymbol{R}^{\mathrm{T}} \bullet \boldsymbol{n}_m \end{bmatrix}}_{m\times 6}, \tag{2}$$

where m represents the number of LiDAR feature points on planar surface patches (hereafter referred to as "planar points"), $\boldsymbol{p}_m^W$ represents the coordinates of points in the world coordinate system,$\hat{}$represents the skew-symmetric matrix of a given vector, used for cross-product operations in 3D space. $\boldsymbol{R}$ represents the state of rotation, and $\boldsymbol{n}$ represents the corresponding normal vector of the plane in the map. We utilize the ikd-tree (Xu et al., 2022) to update the point cloud map and find neighboring points in the map.

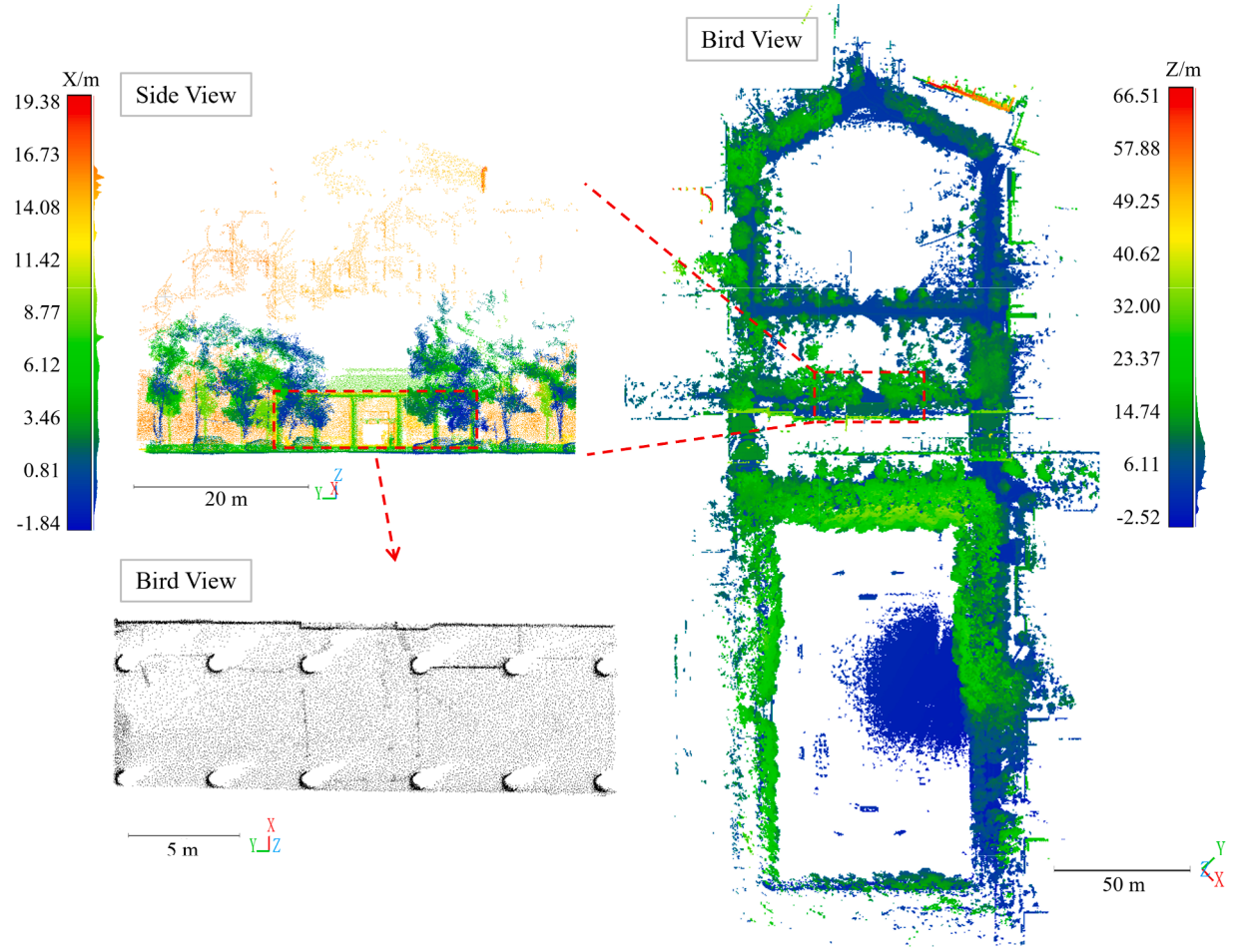
3.3. MC-PGO

PGO is an efficient method for reducing the errors of the front-end. In this section, we aim to improve the performance of PGO by enhancing the accuracy of relative pose constraints between graph nodes through constructing various types of constraints with dedicated strategies.

Considering the single-frame point cloud is sparse, as well as the

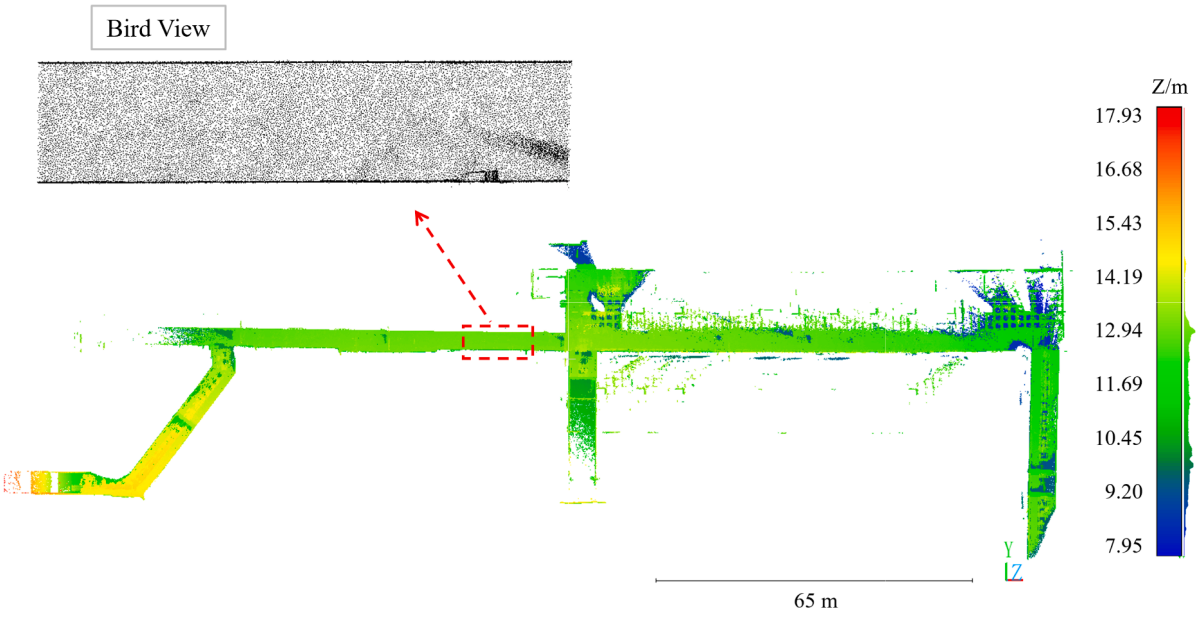


(a) Seq1



(b) Seq2

Fig. 11. The point cloud maps of dataset.



(c) Seq3

Fig. 11. (*continued*).

Table 6
Evaluation of the efficiency (seconds) of different modules in MC-PGO.

Dataset	Modules				
	Submap reconstruction	Submap retrieval	Submap registration	Optimization	Overall
Seq1	33.9	94.3	261.3	0.03	389.53
Seq2	62.3	350.3	188.4	0.02	601.02
Seq3	24.6	10.7	291.1	0.01	326.41

Table 7
Front-end results of using different MDC methods on experiment data.

Dataset	Method	
	Standard ATE RMSE/Max error (m)	DS-MDC (Ours)
Seq1	1.051/1.840	**0.968/1.647**
Seq2	1.068/2.182	**1.006/2.165**
Seq3	2.579/5.298	**2.499/5.194**

Table 8
Ablation results of DA updating strategy on experiment data.

Dataset	Method	
	W/O DA updating ATE RMSE/Max error (m)	With DA updating
Seq1	0.854/1.537	**0.484/1.017**
Seq2	0.842/1.700	**0.693/1.434**
Seq3	2.499/5.194	**2.376/4.850**

restricted horizontal FOV of solid-state LiDAR, we utilize point clouds accumulated over a period of time to construct the submap for backend optimization, which is the smallest unit for PGO. The submap accumulation time t_S will be discussed in the section 5.2.

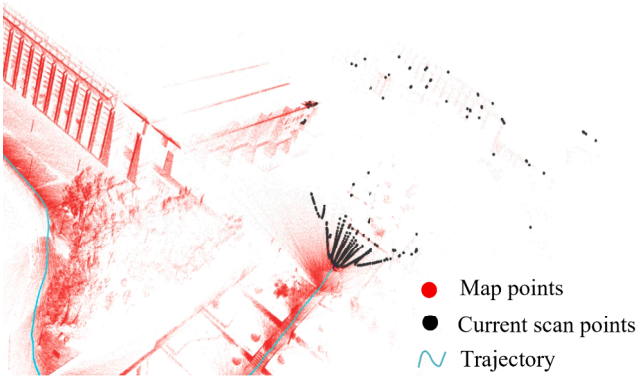
To get the accurate relative pose $\boldsymbol{T}_k^j$ between submap S_j and submap S_k, a simple and efficient ICP method is leveraged. Non-planar points in the submap are filtered out (as shown in Fig. 4) through analyzing the spatial distribution of neighboring points (Demantké et al., 2012). Specifically, voxel-based downsampling is applied to improve the efficiency of the ICP algorithm and to reduce the impact of uneven point density on the selection of neighboring points. For each point, the local neighborhood is defined by a fixed number of nearest neighboring points. We then compute the covariance matrix of these neighboring points and perform principal component analysis (PCA) to obtain the eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$. Points are retained if λ_3 is significantly smaller than both λ_1 and λ_2, indicating a likely planar point. The cost function for registration is as follows:

$$\text{Cost}_{\text{regi}}\left(\boldsymbol{T}_k^j\right) = \sum\nolimits_{i\in\mathscr{I}} \rho(\boldsymbol{N}_i^{\mathrm{T}}\left(\boldsymbol{T}_k^j \boldsymbol{p}_i\right) + d_i), \tag{3}$$

where $\boldsymbol{p}_i$ represents the planar point in the submap S_k, $\boldsymbol{N}_i, d_i$ represents the parameters of the matching plane patch in the submap S_j. The matching will be rejected if the distance between $\boldsymbol{p}_i$ and its closest point in submap S_j exceeds a predefined threshold which help to filter out large outliers. Furthermore, the Huber loss function $\rho()$, is used to strike a balance between the accuracy and robustness of the matching (Huber, 1992).

In order to better leverage the observation information from co-visible regions in space, we introduced three distinct constraint types into the PGO: connective constraints, co-visible constraints, and revisited constraints. If the overlap ratio between adjacent submaps exceeds O_{submap}, a connective constraint will be established. To improve the efficiency of searching for co-visible submaps, the search is limited to the K nearest neighboring submaps of the current submap. Retrieval of revisited submaps can be done using odometry information or place recognition algorithms (Zou et al., 2023). To further enhance the accuracy of revisited constraints, we design the following two strategies:

- Transforming N temporally adjacent submaps into the coordinate system of the current submap to construct a submap group that participates in the registration.



(a) The uneven distribution of current scan points leads to LiDAR degeneracy (map in red, scan in black)

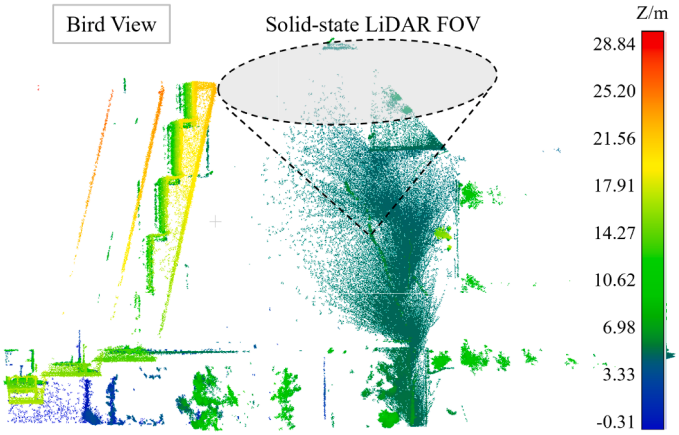


Fig. 12. The LiDAR degenerate case in Seq1 with Livox LiDAR. (a) The uneven distribution of current scan points leads to LiDAR degeneracy (map in red, scan in black). (b) Bird's-eye view of a degeneracy scenario. Due to the open environment and the limited FOV of the solid-state LiDAR, the scan points exhibit an uneven distribution, as shown in (a).

- Adaptively increasing the distance threshold for rejecting point-to-plane matches during the registration.

At last, the cost function for MC-PGO is as follows:

$$\text{Cost}_{\text{PGO}} = w_1 \sum_{(j,k)\in\mathscr{C}} \left\| \left(\boldsymbol{T}_W^j \boldsymbol{T}_k^W \right) \boldsymbol{T}_j^k \right\|^2 + w_2 \sum_{(l,m)\in\mathscr{O}} \left\| (\boldsymbol{T}_W^l \boldsymbol{T}_m^W) \boldsymbol{T}_l^m \right\|^2 + w_3 \sum_{(n,o)\in\mathscr{L}} \rho(\boldsymbol{T}_W^n \boldsymbol{T}_o^W \boldsymbol{T}_n^o), \tag{4}$$

where $\boldsymbol{T}_i^W$ represents the submap poses to be optimized whose initial values of these poses are provided by the odometry. $\boldsymbol{T}_j^k$ represents the relative pose constraint between the S_j and S_k, obtained through pair-wise registration. $\mathscr{C}, \mathscr{O}, \mathscr{L}$ represent the sets of submap indices corresponding to connective, co-visible, and revisited constraints respectively. The weights for these three constraints are denoted as w_1, w_2, w_3 respectively. The value of parameter N was determined by based on the overlap of submap groups.

4. Experiment results

This section provides a detailed description of the experiment data and settings, and evaluates the effectiveness of the proposed method by comparing it to existing representative LIO and back-end optimization methods (i.e., FAST-LIO2, HBA).

4.1. Dataset description

Four diverse sequences datasets shown in Fig. 5 (b) collected in Wuhan, China, were used to verify the validity of DALI-SLAM. Table 1 shows the detailed description of experiment data. As shown in Fig. 5 (c), the helmet-based laser scanning system (HLS) named WHU-Helmet, equipped with a solid-state LiDAR (i.e., Livox Avia) and a MEMS IMU, was used. Three of four data is from WHU-Helmet public dataset (Li et al., 2023a) with cm-level ground-truth trajectory. Specifically, Seq1 from the WHU-Helmet infrastructure dataset was collected on a campus road around a stadium, containing several open areas. Seq2, a subset of the WHU-Helmet roadside dataset, was also collected in a campus road but with plenty of dense trees and dynamic objects. Seq3, a subset of the WHU-Helmet underground space dataset, was collected in an entrance level of subway station with limited geometric features. The remaining data Seq4 was collected in the staircase where geometric features are severely lacking.

4.2. Experiment setup

Baselines. To demonstrate the superiority of the proposed method, we compared the front-end (denoted as DA-LIO) and back-end (denoted as MC-PGO) respectively with FAST-LIO2, CLINS, LeGO-LOAM (only for back-end, denoted as LeGO-LOAM-B), MuCoGraph and HBA. FAST-LIO2 (Xu et al., 2022) is the filter-based SOTA method for LIO. CLINS (Lv et al., 2021) is the optimization-based LIO which utilizes the B-spline

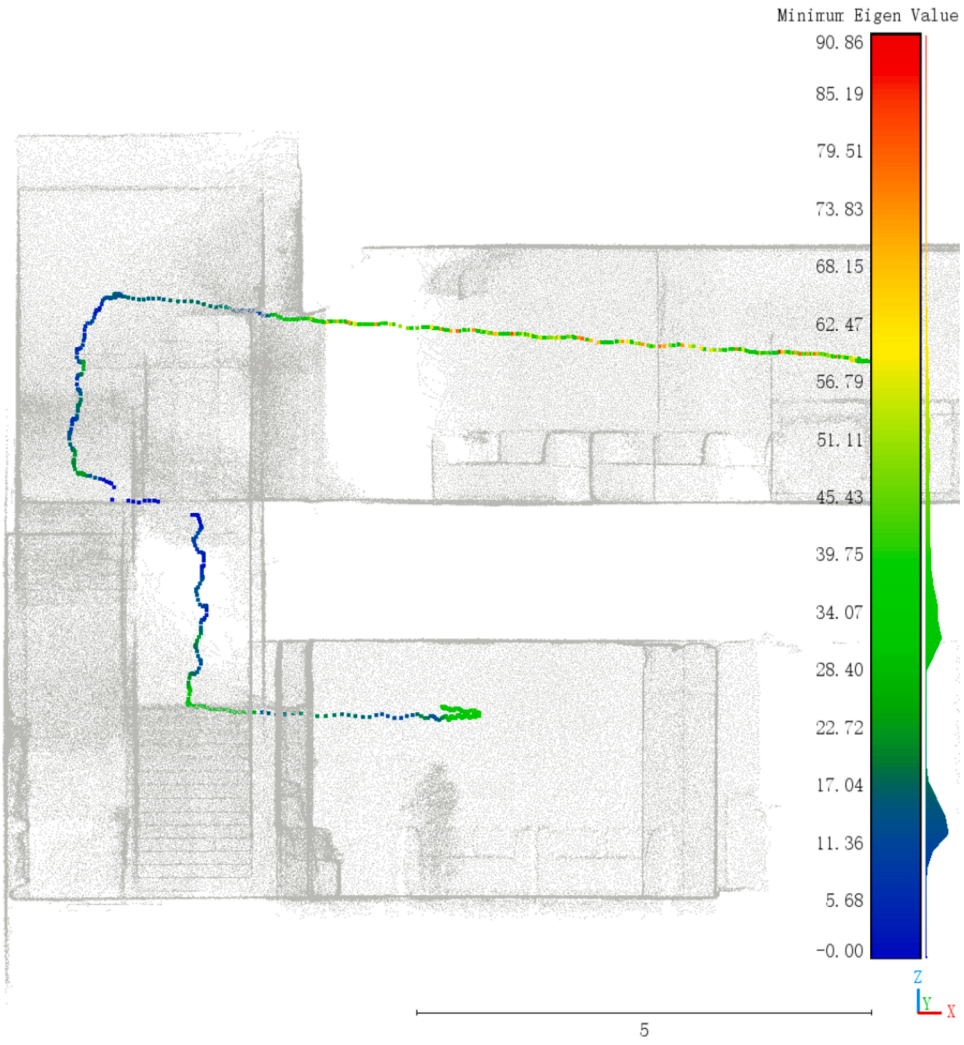


Fig. 13. The minimum eigen values of the system in the staircase data, and the point cloud shown is generated by FAST-LIO2.

Table 9
Ablation results of co-visible constraints on experiment data.

Dataset	Method	
	W/O co-visible constraints ATE RMSE/Max error (m)	With co-visible constraints
Seq1	0.422/1.019	**0.362/0.762**
Seq2	0.652/1.423	**0.629/1.386**
Seq3	2.376/4.850	**1.130/2.622**

Table 10
Back-end results of using different registration methods on experiment data.

Dataset	Method			
	NDT ATE RMSE/Max error (m)	GICP	VGICP	Ours
Seq1	0.477/0.920	7.621/12.312	**0.350**/ 0.842	0.362/ **0.762**
Seq2	1.120/2.326	32.971/ 150.437	0.768/ 1.618	**0.629**/ **1.386**
Seq3	2.392/4.221	36.800/ 78.140	1.259/ 3.458	**1.130**/ **2.622**

trajectory model. LeGO-LOAM (Shan and Englot, 2018) utilizes key frame-based pose graph optimization for back-end. MuCoGraph (Li et al., 2023b) is a multi-scale constraint-enhanced pose graph optimization method specifically designed for vehicle-based point cloud optimization using single-line LiDAR. HBA (X. Liu et al., 2023) is the latest hierarchical BA method for LiDAR consistent mapping. All comparison methods, except for MuCoGraph, are open-sourced. We modified CLINS's IO to successfully load Livox data as it is designed for Velodyne LiDAR originally. To evaluate the front-end and back-end separately, we utilized the format that consists of 6-DOF trajectory and corresponding scans in PCD format for LIO output. In addition, we reimplemented the LeGO-LOAM back-end for off-line processing. The parameters of each method were adjusted multiple times for the fair comparison.

Evaluation criteria. In this paper, we use the ATE (absolute trajectory error) (Sturm et al., 2012) to quantitatively evaluate the effectiveness of DALI-SLAM. The RMSE and maximum error of ATE are computed for evaluation. The normalized trajectory error is calculated by dividing the trajectory error by the trajectory length.

Implementation details. All methods in the experiments were implemented by C++ and tested on a computer with an Intel Xeon(R) E3-1240v5 CPU and 16G RAM. The order of the B-Spline was set to 3, and the knot spacing was set to 50 ms, both of which are determined based on the motion characteristics. The Huber loss parameter for registration was set to 0.05 cm empirically. The co-visible submap search parameter K was set to 10, balancing the efficiency of co-visibility constraint retrieval with optimization accuracy. The selection of the

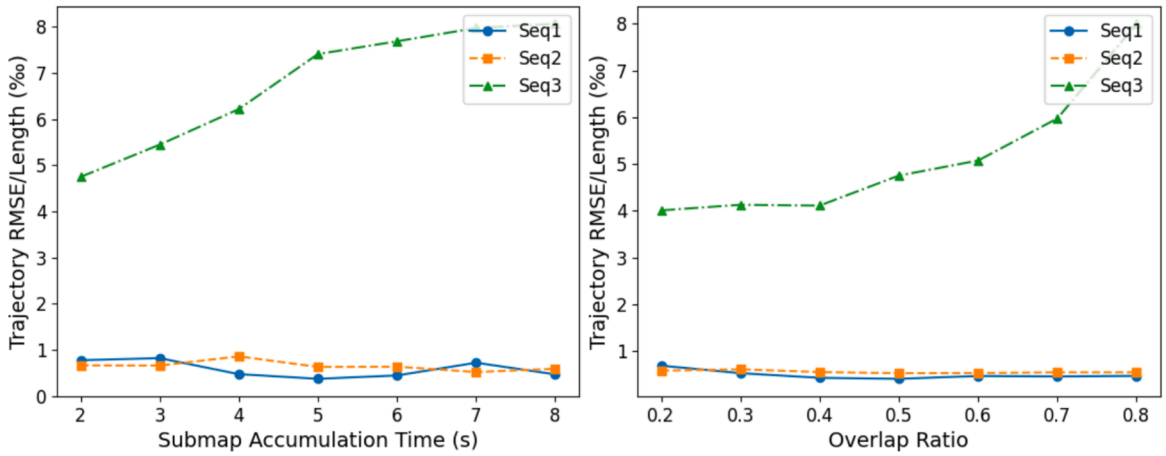


Fig. 14. Trajectory RMSE/length (‰) curve with respect to the submap time and overlap ratio.

pose graph constraint weights is based on empirical observations, suggesting that revisited and co-visible constraints should be assigned greater importance through larger weights. In our experiments, w_1, w_2, w_3 were set to 1, 50, and 10, respectively.

4.3. Quantitative evaluation

Front-end. Table 2 shows the root mean square and max errors of ATE metric for the different LIO methods, and Fig. 6 shows the x-y trajectory of different methods on this dataset. The results indicate that DA-LIO outperforms other methods among 3 sequences. Compared to the filter-based SOTA method FAST-LIO2, DA-LIO achieves an ATE RMSE reduction of 53.9 %, 35.1 % and 7.9 % respectively. It indicates that DA-LIO significantly outperforms FAST-LIO2 due to the proposed module. In addition, we found that CLINS failed on Seq3 because it requires strict still initialization. It shows that there is large improvement room for continuous-time methods. Finally, the ratios of the maximum ATE to the trajectory length were calculated as 0.11 %, 0.12 %, and 1.72 % for Seq1, Seq2, and Seq3, respectively. These results indicate that, the accuracy of Seq3 is significantly lower than that of Seq1 and Seq2 due to the presence of fewer geometric features in Seq3.

Fig. 7 shows the point clouds of different LIO methods on staircase dataset and Fig. 8 shows the z-axis trajectory of different methods. It indicates that FAST-LIO2 experiences large z-axis drift in the middle platform of the staircase, and CLINS experiences significant drift so that the odometry system fails. Due to the narrow horizontal FOV of solid-state LiDAR, the sensor encounters feature degeneracy when it is faced closely to a wall. DA-LIO utilizing degeneracy-aware updating strategy to avoid large drift in this situation.

Table 3 presents the average processing time per scan for each LIO method. It indicates that DA-LIO successfully achieves a balance between accuracy and efficiency.

Back-end. Table 4 shows the root mean square and max errors of ATE metric for the different back-end methods based on the DA-LIO odometry result. According to the trajectory lengths shown in Table 1, the normalized trajectory RMSEs are 0.4 ‰, 0.52 ‰, and 4.01 ‰ for three datasets. The proposed MC-PGO achieves an ATE RMSE reduction of 25.2 %, 9.2 % and 52.4 % on dataset respectively. The results indicate that MC-PGO outperforms other methods among 3 sequences. However, compared with the remaining dataset, the improvement of MC-PGO on Seq2 is relatively limited. The reason may be that the number of co-visible submap constraint is less (as shown in Table 5) due to the lush leaves along the road. Due to a considerable registration error in the revisited submap constraint, the accuracy of LeGO-LOAM-B is notably inferior to the odometry results on Seq1. Furthermore, a revisited submap constraint has not been established for Seq2 and Seq3, resulting in the omission of PGO. MuCoGraph shows limited performance on the three sequences primarily because it constructs pose graph constraints based only on revisited areas. Additionally, MuCoGraph relies on feature descriptors for submap registration without precise registration methods like ICP, resulting in less accurate pose constraints. HBA performs a hierarchical bundle adjustment by optimizing planes; however, spatially adjacent planes may be incorrectly added to the cost function due to voxel association, causing inaccurate optimization. Additionally, the results demonstrate the improvement of HBA in geometric degeneracy scenes like Seq3 is not obvious. MC-PGO selects the pair of submaps through the overlap metric between submaps to mitigate the influences of geometric degeneracy to some extent.

In Fig. 9 and Fig. 10, the point clouds of the revisited regions are shown before and after conducting MC-PGO based on the front-end results. Fig. 11 shows the point cloud map obtained after performing MC-PGO to three different sequences. The results demonstrate an enhancement in geometric consistency after the back-end optimization. Our method is capable of handling various scene types, including structured and structure-less scenes. Table 6 shows the efficiency of different modules in MC-PGO. It indicates that the processing time is comparable to the data collection time. By employing parallel acceleration, the processing time of the submap registration module can be further reduced.

5. Analysis and discussion

5.1. Comparative study

In this section, we conduct a comparative study and analysis of different techniques, including the MDC, the update strategy of the Kalman filter, the co-visible constraint of PGO, and the point cloud registration algorithm utilized in PGO.

Table 7 shows the front-end results of using different MDC methods based on FAST-LIO2, where the term “standard ” in the table refers to the method employed in FAST-LIO2. Our DS-MDC module achieves an ATE RMSE reduction of 7.9 %, 5.8 % and 3.1 % on dataset respectively compared to the existing module in FAST-LIO2.

Table 8 shows the front-end results of using DA updating strategy or not. After using DA updating strategy, the results achieve an ATE RMSE reduction of 43.3 %, 17.7 % and 4.9 % on dataset respectively. Fig. 12 illustrates a representative instance of LiDAR degenerate case in Seq1. The system trajectory is presented in blue, the map in red, and the point cloud captured by the solid-state LiDAR at the current moment in black. Given the limited FOV and the relatively open environment at the current location, a significant portion of the scanned points is primarily distributed along the ground surface and the building facades in the

forward direction. To address this, our method detects LiDAR degeneration and introduces a DA updating strategy into the ESKF to achieve more accurate pose estimation. Fig. 13 illustrates the minimum eigen values of the system, identifying the degenerate places in the staircase.

Table 9 presents the ablation study results on the effectiveness of using co-visible constraints in PGO or not. It demonstrates the benefits of the co-visible constraint, especially for Seq3.

Table 10 shows the back-end results of using different point cloud registration methods on experiment data. To ensure the fairness of the experiment, the same submap downsampling voxel size was used. Multiple parameter adjustments and tests were conducted to select the best result for the comparison methods. The grid size of NDT method (Stoyanov et al., 2012) was set to 0.5 m, and the outlier scale was set to 0.6. In the GICP method (Segal et al., 2009), the number of neighbor points required to calculate covariance was set to 20. As for the VGICP method (Koide et al., 2021), the grid size was set to 1.5 m for Seq1 and Seq2, and 1 m for Seq3. The results demonstrate that, in comparison to the advanced VGICP method, the maximum trajectory error is reduced by 9.5 %, 14.3 %, and 24.2 % across the three sequences, respectively. On the other hand, the GICP method yields the poorest accuracy. This can be attributed to the fact that the GICP method aligns the covariance of the source point cloud and the target point cloud for registration. However, since different downsampling parameters were used for the source and target point clouds in the test, the calculation of covariance is affected by the number of neighbor points. As a result, inconsistent covariance calculations lead to significant errors.

5.2. Parameter sensitivity analysis

Fig. 14 illustrates the curves of trajectory RMSE divided by trajectory length with respect to the submap accumulation time t_S and the overlap ratio parameter O_{submap} respectively while keeping other parameters fixed. t_S was tested in the range of 2 s to 8 s, with a step size of 1 s. O_{submap} was tested in the range of 0.2 to 0.8, with a step size of 0.1. The graph reveals that, based on the current test datasets, the optimal t_S for the indoor scenario (Seq3) is smaller than that for the outdoor environments (Seq1 and Seq2), due to the larger non-rigid errors in the submaps. Additionally, for Seq3, it is necessary to set a small value for the O_{submap} to ensure the construction of sufficient co-visible constraints. Furthermore, in Seq1 and Seq2, the algorithm appears to be less sensitive to these two parameters than in Seq3.

5.3. Deficiency and future work

The experiment results demonstrate that the proposed method integrates the degeneracy-aware LIO and multi-constraint PGO to achieve robust and accurate localization and mapping results in various indoor and outdoor scenes. However, our method still has some limitations that warrant further investigation. Firstly, if the duration of complete LiDAR feature degeneracy is too long, our method will fail as the drift of system states derived from IMU integration accumulates rapidly over short time periods. Secondly, according to our experiment and existing research (Tuna et al., 2023), the threshold λ_D for detection LiDAR degeneracy needs to be adjusted based on the scene and the specific LiDAR used. Lastly, the non-rigid errors of submaps are ignored in our method through selecting suitable submap accumulation time in different scenes.

In the future work, we will conduct research in following aspects to further improve the robustness and accuracy of the LI-SLAM system. Firstly, to improve the robustness in extreme LiDAR degeneracy scenes, we will investigate leveraging the Doppler velocity information (Wu et al., 2022b) provided by FMCW (Frequency Modulated Continuous Wave) LiDAR in the front-end part. Alternatively, dynamic initialization should be explored to recover the system from failure when both LiDAR odometry and IMU experience significant drifts. Secondly, to enable adaptive LiDAR-IMU fusion, we will explore the use of deep neural networks (Nubert et al., 2022) to detect LiDAR degeneracy. Finally, we plan to study integrating raw LiDAR matching constraints with relative pose constraints into the back-end optimization efficiently. In addition, we will explore methods for removing dynamic object points or tracking dynamic objects in the environment, enabling the construction of dynamic object-aware maps. Moreover, we will investigate the impact of transparent surfaces on the map, with the goal of reducing specular noise and minimizing their effects on localization and navigation.

6. Conclusion

In this paper, we propose a LiDAR-inertial SLAM method consisting of DA-LIO and MC-PGO, which effectively overcomes challenges posed by inaccurate motion distortion estimation and pose graph constraints, and frequent feature degeneracy. Specifically, a novel dual B-spline-based motion distortion correction method which utilizes the continuous-time trajectory model to update integrated IMU poses is proposed to enhance the correction accuracy. Additionally, degeneracy-aware Kalman update strategy makes the odometry believe raw non-degenerate information, thereby improving the system accuracy. Finally, a robust and simple ICP with dedicated strategy is utilized to accurately build three types of submap constraints in PGO. We have validated the effectiveness of the proposed method on an HLS dataset. The experiment demonstrates that the normalized trajectory RMSEs of DALI-SLAM are 0.4 ‰, 0.52 ‰, and 4.01 ‰ respectively on the test data. Furthermore, ablation experiments verify the rationality and effectiveness of the core modules of our methods. In the future, we aim to utilize Doppler measurements in the odometry part, and integrate the relative pose constraints and the raw feature constraints in the back-end to further improve the accuracy and robustness of LI-SLAM system.

CRediT authorship contribution statement

Weitong Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Data curation, Conceptualization. **Chi Chen:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Bisheng Yang:** Supervision, Resources, Funding acquisition, Conceptualization. **Xianghong Zou:** Writing – review & editing, Methodology, Data curation. **Fuxun Liang:** Writing – review & editing, Methodology. **Yuhang Xu:** Writing – review & editing. **Xiufeng He:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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